

Censorship as Bitcoin Grows

A Powerful Dummy's guide to censoring Bitcoin

Micah Warren

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Department of Mathematics
University of Oregon

Intro

- Salient topic: North Korean hackers, CSAM, pig-butcherer, hospital ransomware, Iran

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- happens on other chains

- Feather Forking (is it enough to control a non trivial minority of hashrate?)

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- Proposal to use leverage provided by economic nodes directly .(see James talk)



Classical Approach: Feather Forking

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- $\implies$ 4% risk to miner for including a censored transaction

- Discussed on Bitcoin talk as far back as 2013

Feather Forking

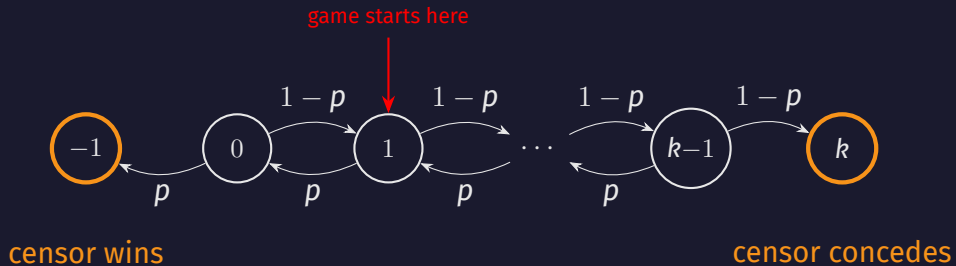
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- Discussed at length in Hendrickson and Warmke 2026

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- I roughly agree
- I use different models - fix hashrate of each agent, run the game and they decide their strategy

Random Walk Model



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- This is a finite difference equation (like discrete 2nd order ODE)
- Add a boundary condition $W(-1) = 1$ and $W(k) = 0$ and this becomes solvable

$$W(x) = \frac{\left(\frac{p}{1-p}\right)^k - \left(\frac{p}{1-p}\right)^x}{\left(\frac{p}{1-p}\right)^k - \left(\frac{p}{1-p}\right)^{-1}} \quad (1)$$

Probability of a Successful Reorg

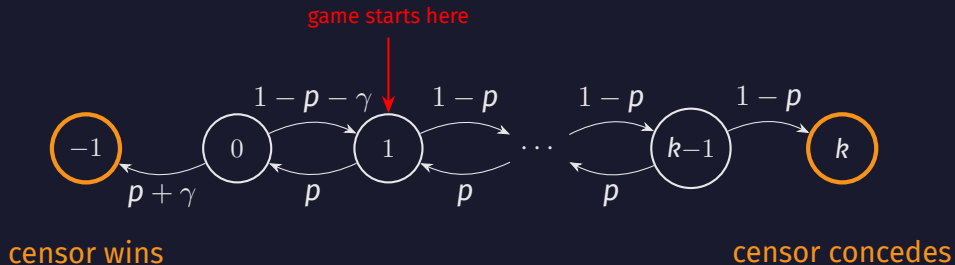
p	$W(1), k = 2$	$W(1), k = 5$
0.10	0.0110	0.0123
0.20	0.0476	0.0623
0.30	0.1139	0.1786
0.45	0.2691	0.5278
0.60	0.4737	0.8797

Break-even Fee f^*

Minimum fee required to make mining the noncompliant block rational:

p	$f^*, k = 2$	$f^*, k = 5$
0.10	0.0112	0.0128
0.20	0.0512	0.0717
0.30	0.1335	0.2561
0.45	0.3902	1.5711
0.60	0.9711	10.9032

Random Walk Model 2 some miners rationally join censor



- Costly to miners - may require subsidies

Feather Forking Analysis

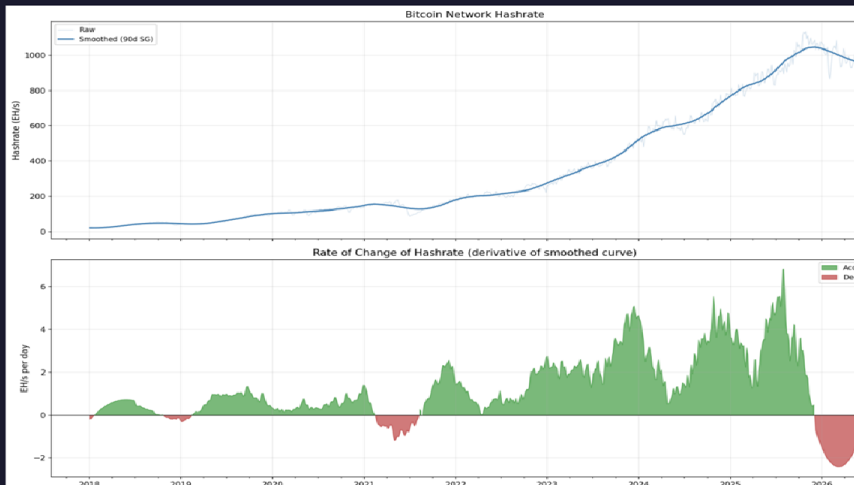
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- Unlikely to work all the time
- Drawn out races could be counterproductive

Fortifying Feather Forks

Hashrate stockpile : Napkin Math



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- Energy draw: 8–12 GW $\approx 2\%$ of US grid — not trivial, must be geographically distributed
- At $\$0.10/\text{kWh}$: $\$1\text{M/hr}$, $\$24\text{M/day}$ — expensive sustained, **cheap if used only for hours**

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- As subsidy decreases and price doesn't double, could become more feasible

Hash rental and pool operating

- economically motivated miners are economically motivated

Running a pool

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- Calculus entirely depends on how many miners can be wooed by higher payouts

- **Full Pay Per Share** - pool pays miner for partial proofs of work

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- FPPS pools are attractive during race, miners will seek out highest paying pool and point there

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- Misaligned incentives during race
- Very hard to detect conclusively

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- This still does not prevent some block withholding

Big Hard Problems With Soft Answers

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- No more happy path for stakeholders in general

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- What possible responses are on the table?

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- In response to disruption?

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- options exist – INVALIDATEBLOCK, checkpointing, temporary sentinels, "bunker mode"

Big question: who ultimately controls Bitcoin

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- Significant Economic nodes on incompatible chains $\implies$ existential crisis
- Economic nodes committing to a side $\implies$ game of chicken
- Domiciled institutions are committed to your side (no choice) $\implies$ the rest of the network has no choice
- (Willingness to blow everything up rather than accept censorial overlords is a real and available very Thomas Schelling-coded option, if economic power is available.)

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- If you have economic control... . why not just wield that directly?

Tainting: How to leverage institutional adoption

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- Bitcoin's UTXO structure allows it
- Softer touch than reorging

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- Institutions must certify they didn't give anything of value when receiving the tainted coins

Objective: Two coins

institutional demand is the load-bearing element, quietly do all the work here –
and that changes everything

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- Dusting – If users choose updated wallets, not a problem
- Compliance burden – Yes - not deep, but wide

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- Use DeFi (Thorchain, for example) to anchor market prices early

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- Stage 4: Inch retroactive, apply market pressure to exchanges to do due diligence

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- Permanent: bad actors can't just RBF out of it
- End game is a button to freeze (at our discretion) any UTXO we don't like - very useful tool!

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- Unclear what happens if other jurisdictions try it
- May cause parties who don't like this power wielded to avoid Bitcoin entirely (works too well! giving US too much arbitrary power)

Thank you!